

Calculus I Lesson 47B

$$\text{Formula A: } \int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$$

$$\text{Formula B: } \int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$$

$$\text{Formula C: } \int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \operatorname{arcsec}\left(\frac{|u|}{a}\right) + C$$

Question 1:

$$\int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \arcsin\left(\frac{|u|}{a}\right) + C$$

Find $\int \frac{2}{x\sqrt{16x^2 - 9}} dx$.

Hint: Let $a^2 = 9 \Rightarrow a = 3$

Let $u^2 = 16x^2 \Rightarrow u = 4 \cdot x$ and $x = \frac{1}{4}u$

Hence, $\frac{du}{dx} = 4 \Rightarrow du = 4 \cdot dx$ and $\frac{1}{4} du = dx$

a) Write the integral $\int \frac{2}{x\sqrt{16x^2 - 9}} dx$ in terms of u : $\int \frac{2}{x\sqrt{16x^2 - 9}} dx = 2 \cdot \int \frac{1}{\frac{1}{4}u\sqrt{u^2 - a^2}} \cdot \frac{1}{4} du = \underline{\hspace{2cm}} ?$

b) $\int \frac{2}{x\sqrt{16x^2 - 9}} dx = 2 \cdot \int \frac{1}{x\sqrt{16x^2 - 9}} dx = \underline{\hspace{2cm}} ?$

$$\int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \operatorname{arcsec} \left(\frac{|u|}{a} \right) + C$$

Question 2:

Find $\int \frac{1}{x\sqrt{9x^2 - 1}} dx$

Let $a^2 = 1 \Rightarrow a = 1$

Let $u^2 = 9x^2 \Rightarrow u = 3x \Rightarrow x = \frac{1}{3}u$

Hence, $\frac{du}{dx} = 3 \Rightarrow \frac{1}{3} du = dx$

a) Write the integral $\int \frac{1}{x \cdot \sqrt{9x^2 - 1}} dx$ in terms of u : $\int \frac{1}{x \cdot \sqrt{9x^2 - 1}} dx = \int \frac{1}{\frac{1}{3}u \cdot \sqrt{u^2 - 1}} \cdot \frac{1}{3} du = \underline{\hspace{2cm}}$

b) $\int \frac{1}{x \cdot \sqrt{9x^2 - 1}} dx = \underline{\hspace{2cm}}$

Question 3:

$$\int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \operatorname{arcsec} \left(\frac{|u|}{a} \right) + C$$

Find $\int \frac{5}{4x\sqrt{25x^2 - 36}} dx$.

Hint: Let $a^2 = 36 \Rightarrow a = 6$

Let $u^2 = 25x^2 \Rightarrow u = 5 \cdot x$ and $x = \frac{1}{5}u$

Hence, $\frac{du}{dx} = 5 \Rightarrow du = 5 \cdot dx$ and $\frac{1}{5}du = dx$

a) Write the integral $\int \frac{5}{4x\sqrt{25x^2 - 36}} dx$ in terms of u : $\int \frac{5}{4x\sqrt{25x^2 - 36}} dx = \frac{5}{4} \cdot \int \frac{1}{\frac{1}{5}u\sqrt{u^2 - a^2}} \cdot \frac{1}{5} du = \underline{\hspace{2cm}} ?$

b) $\int \frac{5}{4x\sqrt{25x^2 - 36}} dx = \underline{\hspace{2cm}} ?$

Question 4:

$$\int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \operatorname{arcsec} \left(\frac{|u|}{a} \right) + C$$

Find $\int \frac{3}{5x\sqrt{5x^2 - 17}} dx$.

Hint: Let $a^2 = 17 \Rightarrow a = \sqrt{17}$

Let $u^2 = 5x^2 \Rightarrow u = \sqrt{5} \cdot x$ and $x = \frac{1}{\sqrt{5}} u$

Hence, $\frac{du}{dx} = \sqrt{5} \Rightarrow du = \sqrt{5} \cdot dx$ and $\frac{1}{\sqrt{5}} du = dx$

a) Write the integral $\int \frac{3}{5x\sqrt{5x^2 - 17}} dx$ in terms of u : $\int \frac{3}{5x\sqrt{5x^2 - 17}} dx = \frac{3}{5} \cdot \int \frac{1}{\frac{1}{\sqrt{5}} u \sqrt{u^2 - a^2}} \cdot \frac{1}{\sqrt{5}} du = \underline{\hspace{2cm}}$

b) $\int \frac{3}{5x\sqrt{5x^2 - 17}} dx = \underline{\hspace{2cm}}$

$$\int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \operatorname{arcsec} \left(\frac{|u|}{a} \right) + C$$

Question 5:

Find $\int \frac{7}{12x\sqrt{14x^2 - 7}} dx$.

Hint: Let $a^2 = 7 \Rightarrow a = \sqrt{7}$

Let $u^2 = 14x^2 \Rightarrow u = \sqrt{14} \cdot x$ and $x = \frac{1}{\sqrt{14}} u$

Hence, $\frac{du}{dx} = \sqrt{14} \Rightarrow du = \sqrt{14} \cdot dx$ and $\frac{1}{\sqrt{14}} du = dx$

a) Write the integral $\int \frac{7}{12x\sqrt{14x^2 - 7}} dx$ in terms of u : $\int \frac{7}{12x\sqrt{14x^2 - 7}} dx = \frac{7}{12} \cdot \int \frac{1}{\frac{1}{\sqrt{14}} u \sqrt{u^2 - a^2}} \cdot \frac{1}{\sqrt{14}} du = \underline{\hspace{2cm}} ?$

b) $\int \frac{7}{12x\sqrt{14x^2 - 7}} dx = \underline{\hspace{2cm}} ?$

